

SAHIL SOMYANI

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portfolio: sahil.page | <https://github.com/sahil007-ai>
Data Scientist / Machine Learning Specialist / Python Developer

PROFESSIONAL SUMMARY

Data Scientist skilled in Python, Machine Learning, and Statistical Modeling. Built end-to-end ML pipelines for predictive risk and churn analytics, translating data into actionable business insights.

EDUCATION

JD College of Engineering

Bachelor of Technology in Artificial Intelligence

Nagpur, India

Jan 2024 – Sept 2028 (Expected)

– **CGPA:** 8.35 / 10.0

– **Relevant Coursework:** Probability & Statistics, Linear Algebra, Data Structures, DBMS, Machine Learning, Data Visualization

TECHNICAL SKILLS

Languages: Python (Advanced), SQL, C++

Data Science: Scikit-learn, Pandas, NumPy, Matplotlib, Seaborn, TensorFlow, PyTorch, OpenCV, Streamlit

Core Competencies: Exploratory Data Analysis (EDA), Statistical Inference, Time-Series Forecasting, Feature Engineering

PROJECTS

Predictive Adulteration System | *Machine Learning Pipeline*

- Architected a predictive classification model to identify high-risk food categories using **FSSAI surveillance data**, building an ETL pipeline for raw data ingestion and model deployment.
- Engineered domain-specific features, integrating seasonal constraints and mapping algorithms to optimize a **Scikit-learn** pipeline.
- Deployed an interactive JavaScript/JSON **Data Visualization Heatmap** and simulated inference tool for seamless public access and health awareness monitoring.

Customer Churn & Behavioral Analytics | *Predictive Modeling*

- Spearheaded attrition risk analysis on 10,000+ customer records using **XGBoost**, achieving an **F1-score of 0.88** by resolving severe class imbalances via **SMOTE** oversampling.
- Executed **Feature Importance Analysis** utilizing SHAP values to isolate key churn drivers (e.g., contract type, tenure), delivering data-backed retention strategies.
- Developed an automated reporting dashboard using **Streamlit** to visualize customer segmentation clusters and highlight predicted high-risk profiles for targeted marketing.

Focus Guard (AI Pomodoro Timer) | *1st Place, Mini Project Exhibition 2025*

- Engineered a real-time computer vision desktop app using **OpenCV** and **MediaPipe** to monitor user engagement via head pose estimation and Eye Aspect Ratio (EAR) for drowsiness detection.
- Secured **1st Position** by implementing an adaptive machine learning algorithm that dynamically adjusts user work session durations based on actual focus patterns and visual feedback.

CERTIFICATIONS

- **Machine Learning Specialization** – Kaggle (Comprehensive Data Analysis, Feature Engineering & Modeling)

VOLUNTEERING & LEADERSHIP

JD College of Engineering

2025 – Present

Resume Workshop Facilitator & Peer Mentor

- Organized and hosted a technical resume-building workshop for university peers, teaching ATS-optimization strategies that resulted in a **90% success rate** of students generating professional resumes.

National Service Scheme (NSS)

2024 – Present

Education Volunteer & STEM Mentor

- Mentored underprivileged students by simplifying complex STEM logic into accessible concepts, demonstrating a strong aptitude for communicating technical subjects to non-technical audiences.